Omics-1

Milestone 4: GSEA

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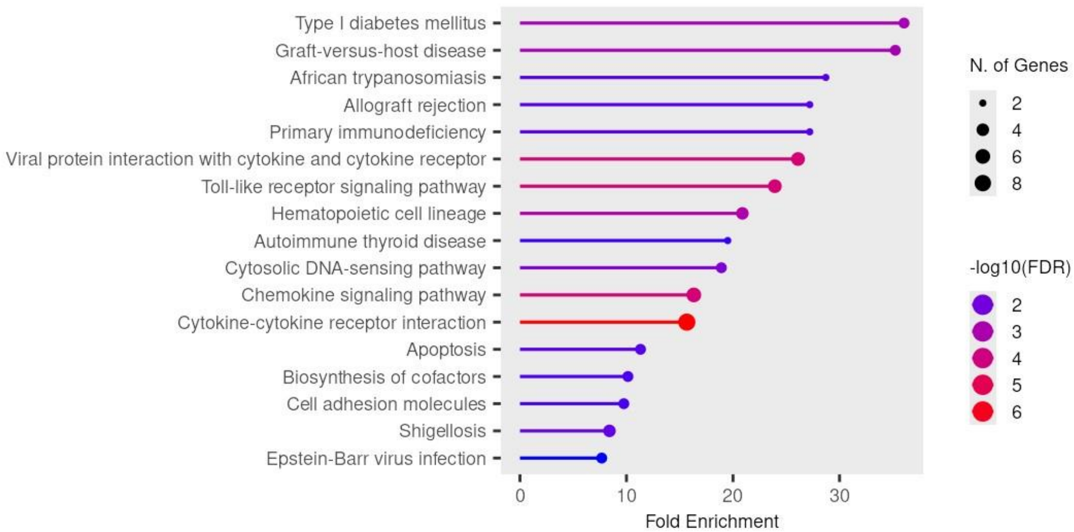
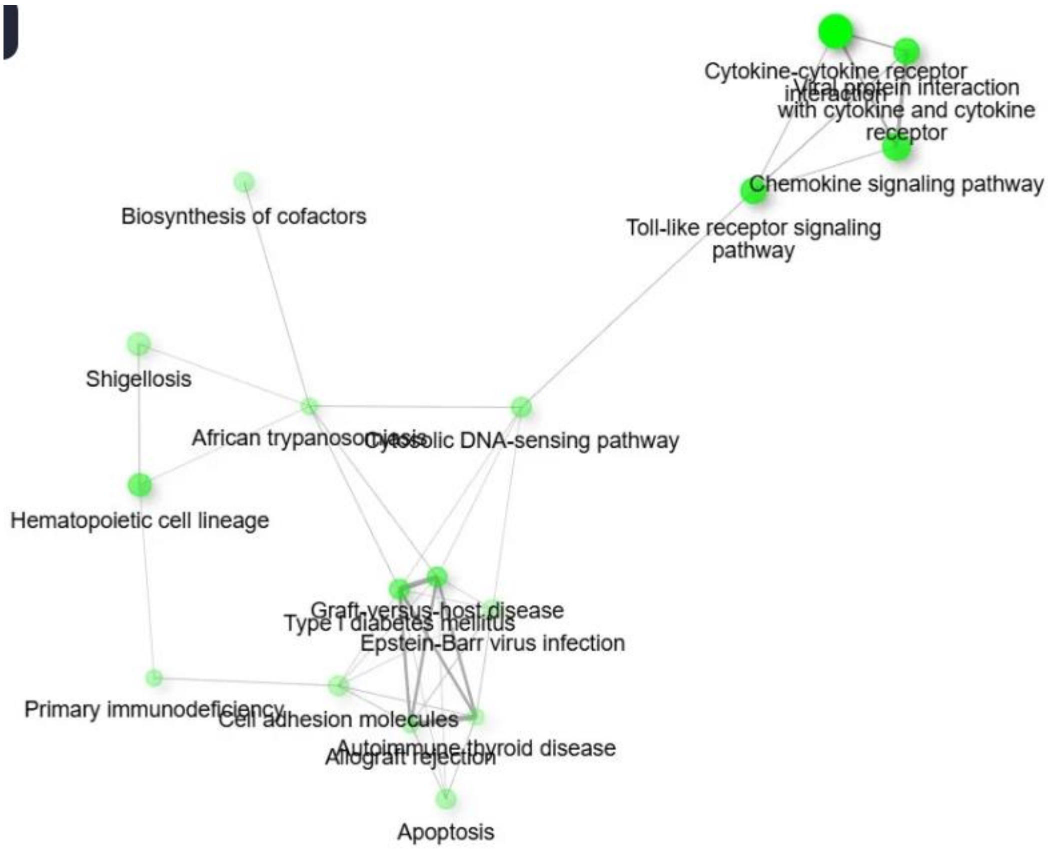
Figure 1 — KEGG Pathway Enrichment Chart (Lollipop Plot)



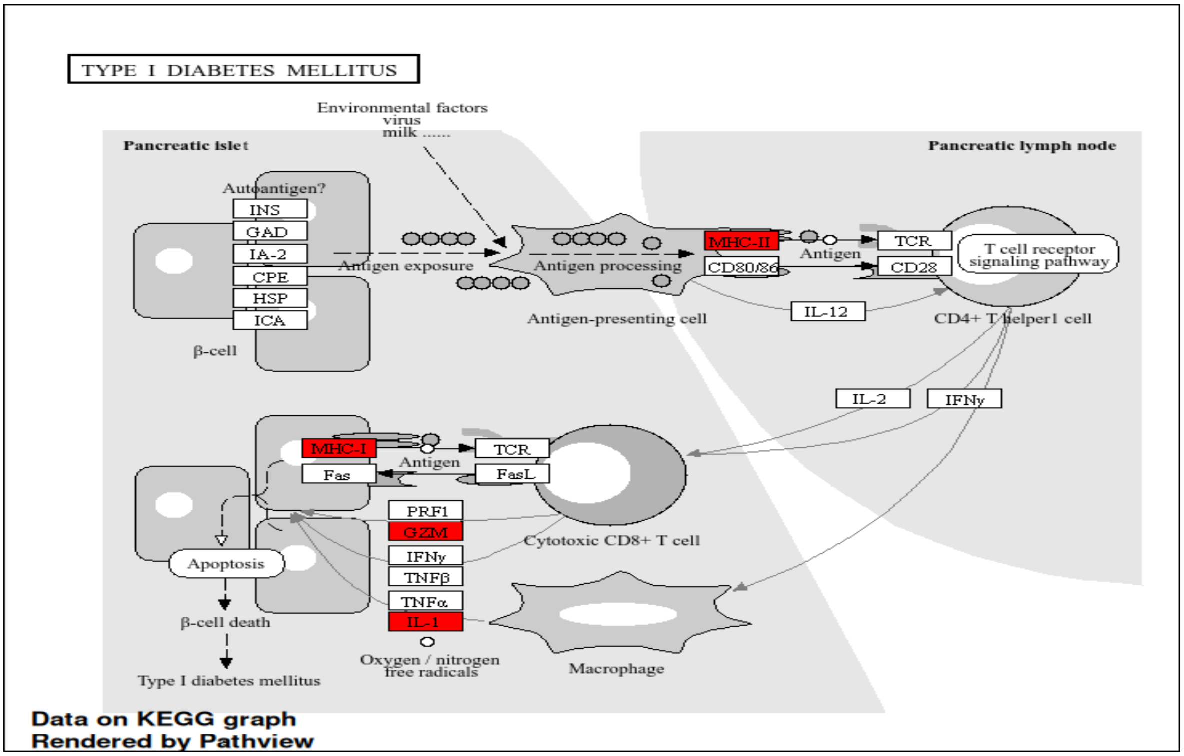
This chart shows the biological pathways most activated in spinal TB compared to disc degeneration. The longer the bar, the more enriched the pathway. Almost all top pathways are immune-related, cytokine signaling, Toll-like receptor signaling, and immune cell interactions, which makes sense because spinal TB is fundamentally an infectious disease where the immune system is working overdrive to fight M. tuberculosis. The most statistically significant pathway (red dot) is cytokine-cytokine receptor interaction, reflecting the intense inflammatory environment TB creates in spinal tissue.

Figure 2 — Pathway Network (Enrichment Map)



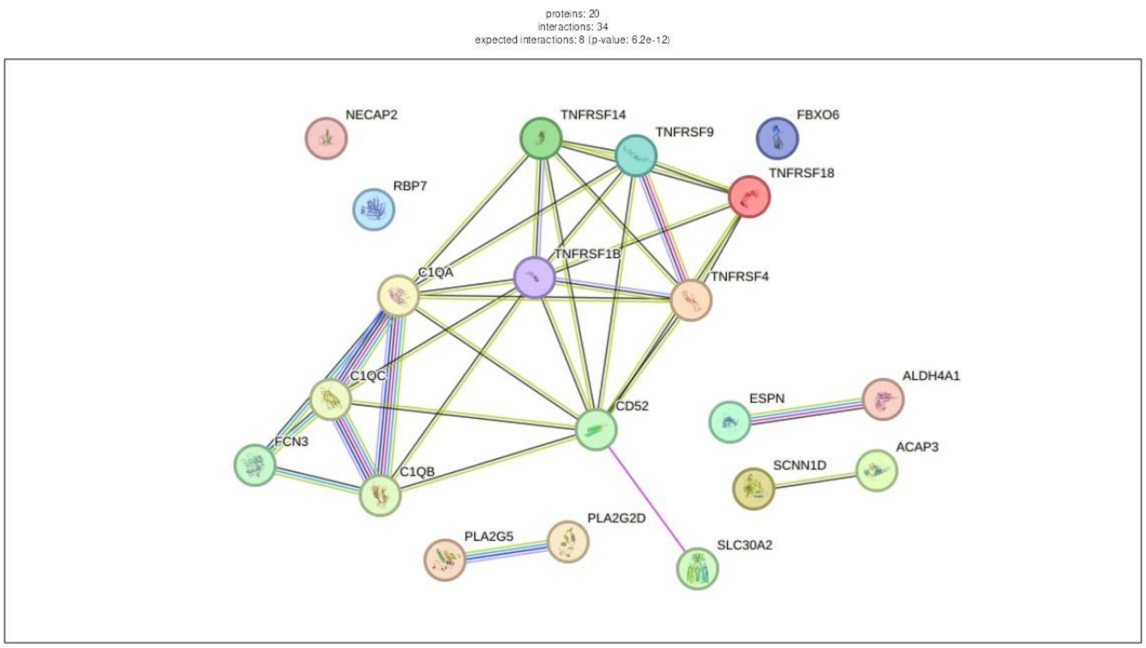
This network shows how the enriched pathways are connected to each other through shared genes. Two clusters are visible, one grouping autoimmune and immune-related diseases, and another grouping cytokine and receptor signaling pathways. This tells us that in spinal TB, two things are happening simultaneously: the body is mounting a broad immune response and driving intense local inflammation at the site of infection in the spine.

Figure 3 — KEGG Pathway Map



The red boxes are genes from the spinal TB dataset that overlap with this pathway. TB appears here not because it causes diabetes, but because both conditions use the same immune destruction machinery, cytotoxic T cells, MHC-I, IFN-γ, TNF, and IL-1. In TB, these molecules are directed at destroying M. tuberculosis-infected cells in the spine, which also causes the collateral tissue damage responsible for vertebral destruction in Pott's disease.

Figure 4 — Protein-Protein Interaction (PPI) Network



This network shows how the top 20 differentially expressed proteins interact with each other. Two main clusters stand out. The TNFRSF proteins (TNF receptors) are central to granuloma formation, the immune structure the body builds to wall off TB bacteria in the spine. The complement proteins (C1QA, C1QB, C1QC, FCN3) represent the innate immune system's first-line attack on the bacteria. Together, these two hubs capture the core of how the immune system responds to and is dysregulated by spinal tuberculosis.